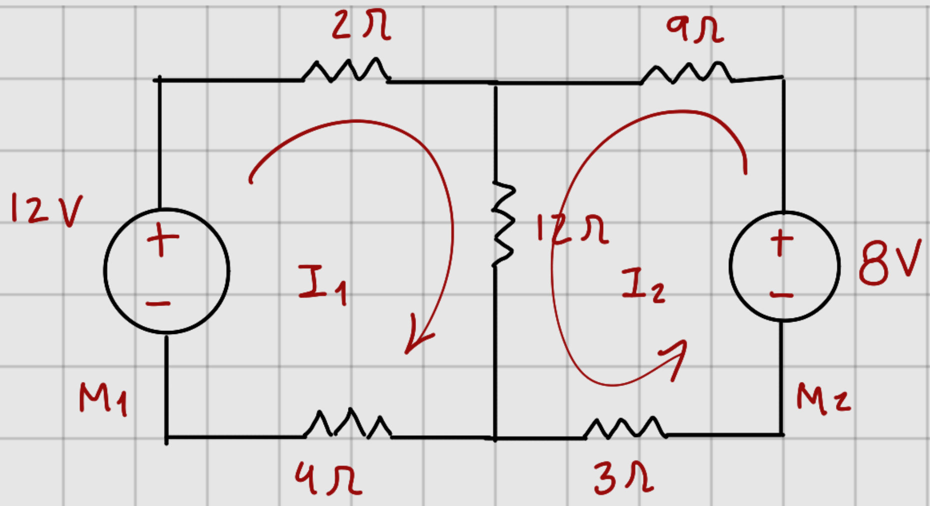


$M_1)\ 2\Omega\, i_1 + 12\Omega\,(i_1+i_2) + 4\Omega\, i_1 = 12V$

$2\Omega\, i_1 + 12\Omega\, i_1 + 12\Omega\, i_2 + 4\Omega\, i_1 = 12V$

$18\Omega\, i_1 + 12\Omega\, i_2 = 12V$

$M_2) = 9\Omega\, i_2 + 3\Omega\, i_2 + 12\Omega\,(i_2+i_1) = 8V$

$9\Omega\, i_2 + 3\Omega\, i_2 + 12\Omega\, i_2 + 12\Omega\, i_1 = 8V$

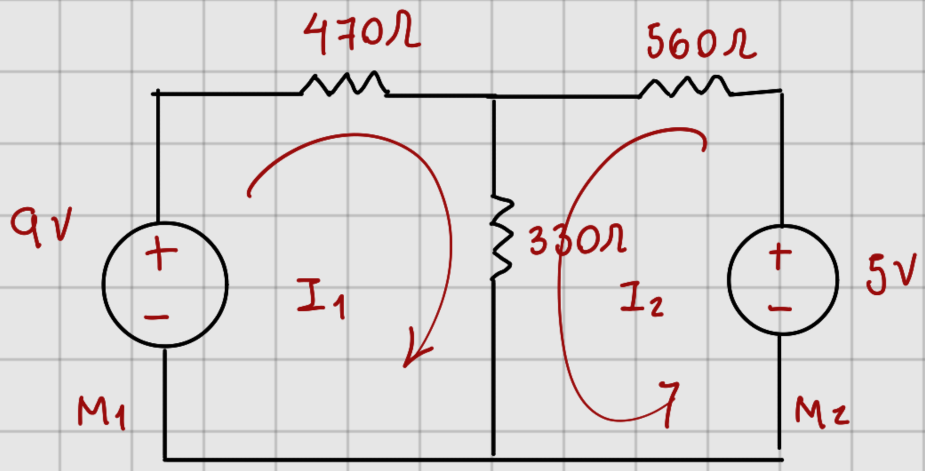
$24\Omega\, i_2 + 12\Omega\, i_1 = 8V$

$18\Omega\, i_1 + 12\Omega\, i_2 = 12V$
$12\Omega\, i_2 + 24\Omega\, i_2 = 8V$

$18\Omega\, i_1 - 12\Omega\, i_2 = 12V$

$i_1 = 0.666\ A$

$i_2 = 0$



$$470i_1 + 330(i_1 + i_2) = 9V$$

$$470i_1 + 330i_1 + 330i_2 = 9V$$

$$800i_1 + 330i_2 = 9V$$

$$560i_2 + 330(i_2 + i_1) = 5V$$

$$560i_2 + 330i_2 + 330i_1 = 5V$$

$$890i_2 + 330i_1 = 5V$$

$$800i_1 + 330i_2 = 9V$$
$$330i_1 + 890i_2 = 5V$$

$$i_1 = 10.5mA$$

$$i_2 = 1.7mA$$